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## Alpha heating and avalanche effect simulations for low density proton-boron fusion plasma

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**ABSTRACT:** The initial interest in p-<sup>11</sup>B fusion, which produces three (3) alpha particles with total energy of 8.7 MeV, was regained the last few years, due to the important experimental measurements on alpha particle production and theoretical and numerical investigations. The re-evaluation of proton-boron fusion, as an important vector for “aneutronic” energy production, is based on the consideration of the “chain reactions alpha heating effect and the related avalanche effect”, as the important process, for the increase of the fusion species (p, <sup>11</sup>B) temperatures, to temperatures corresponding to the optimum p-<sup>11</sup>B fusion cross sections. Investigation of ignition and self-sustained conditions for low density ( $\sim 10^{20} \text{ m}^{-3}$ ) proton-boron fusion plasmas is an interesting study topic for application in magnetic confinement. We use a multi-fluid, global particle and energy balance code, describing the temporal evolution of the physical parameters of the fusion medium, as a function of the initial conditions of density and temperature of the fusion species (p, <sup>11</sup>B). For the establishment of the distinct contribution role of the avalanche effect in proton-boron fusion, we explore cases of low ( $< 100 \text{ keV}$ ) and high ( $> 100 \text{ keV}$ ) initial p-<sup>11</sup>B medium temperatures. For these temperature cases, density ratios  $n_p/n_B > 1$  between the fusion species (p, <sup>11</sup>B) are considered, for the optimization of Bremsstrahlung radiation losses. Simulations using the multi-fluid code in a low density p-<sup>11</sup>B medium with  $n_p/n_B > 1$ , enable the evaluation of the initial temperatures range, which is necessary for the temporal increase of the alpha density, the rapid improvement in the p-<sup>11</sup>B fusion reaction rate (RR), the temperature rise of the fusion species (p, <sup>11</sup>B) and the achievement of higher than 1  $Q$  ( $P_f/P_{\text{Brems}}$ ) values, due to the manifestation of the avalanche effect. In the case of high initial fusion medium temperatures in the range:  $150 \text{ keV} \leq T_{\text{in}} \leq 350 \text{ keV}$ , the chain reactions alpha heating effect and the related avalanche effect contribute to the maintenance and the rise of the fusion species (p, <sup>11</sup>B) temperature to high values, for ignition and self-sustained fusion ( $Q > 1$ ).

**KEYWORDS:** Plasma generation (laser-produced, RF, x ray-produced); Plasma diagnostics - charged-particle spectroscopy

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## 1 Introduction

The recent gain results from inertial confinement fusion at the National Ignition Facility (NIF) [1, 2], confirm that the D-T reaction is the most feasible fusion reaction for big scale energy production on planet Earth, due to: i) Its highest cross section value (5.0 barn) at the lowest center-of-mass energy (100 keV) and ii) The requirement of ignition temperatures in the low energy range between 4 keV and 10 keV [3]. However, due to the emission of an intense neutron flux (1 neutron of 14.1 MeV energy per fusion reaction), D-T inertial or magnetic confinement fusion reactors induce radioactivity, via the activation of the chamber material. Thus, a thick blanket surrounding the plasma chamber is required, in combination with an advanced technological breeder for Tritium (T) production and a high efficiency turbine system for the conversion of the thermal energy into electricity. The three (3) latter facts increase the D-T reactor complexity and entail adverse economic effects on clean fusion energy production [4]. For the aforementioned reasons, the scientific community turns towards charged-particle neutron-free fusion reactions with significant cross section and energy release. In this context, p-<sup>11</sup>B fusion reaction is the most promising candidate for clean energy production [5–7], as: i) It is the only neutron-free fusion reaction ( $< 1\%$  of the energy in neutrons) [5], ii) It produces three (3) iso-energetic charged alpha particles, with 8.7 MeV total energy, that can directly be converted into electricity, without passing through a thermodynamic cycle [4, 8], iii) It doesn't necessitate the use of breeding technologies for the production of its components (p, <sup>11</sup>B) [8, 9].

However, there is a widely shared opinion that the p-<sup>11</sup>B fuel cycle is not an appropriate option for clean energy production, due to its low reactivity [ $\langle\sigma v\rangle$ ], for energies lower than 250 keV [4, 8, 10].

This is a just misconception, as the  $p\text{-}^{11}\text{B}$  fusion reactivity is only a factor of few lower than that of D-T nuclear fusion reaction [4, 9, 11]. The resonances of major interest for  $p\text{-}^{11}\text{B}$  fusion are located at the center-of-mass energies of 163 keV (with a width of just 5 keV) and 675 keV (with a width of 300 keV) [8]. At the 675 keV resonance, the  $p\text{-}^{11}\text{B}$  fusion cross section reaches its maximum value, 1.2 barn [4, 10–12], very comparable to that of D-T nuclear fusion reaction ( $\sim 5$  barn), at the center-of-mass energy of  $\sim 65$  keV [11].

A major drawback of the  $p\text{-}^{11}\text{B}$  nuclear fusion reaction is that, in order to reach fusion ignition conditions and high reactivity values (close to that of D-T), ion temperatures between 250 keV and 500 keV are needed [3, 4, 10, 13–15]. At these high temperatures, which are an important and hard challenge to be achieved with the currently existing laser-based technologies, as the few kJ L4 Anton laser system in Czech Republic [16] or the few J ZEUS-US Laser Facility [17], the electrons become quite hot and lead to the emission of intense Bremsstrahlung radiation [4, 8, 10]. According to D.C. Moreau [10], who was the first one to examine the viability of the  $p\text{-}^{11}\text{B}$  fuel for thermonuclear fusion in the middle of 1970s, the approximation of break-even conditions ( $Q = P_{\text{fus}}/P_{\text{Brems}} = 1$ ) is feasible, if the ratio of  $T_i/T_e \sim 2$  ( $T_i$ : Ion temperature,  $T_e$ : electron temperature) is assumed in combination with recovery techniques, for the optimization of Bremsstrahlung and/or synchrotron radiation losses. Significant progress in the field of  $p\text{-}^{11}\text{B}$  fusion was achieved almost 30 years later, after the important experimental results of [18–21], and [22], who measured  $10^3$   $p\text{-}^{11}\text{B}$  fusion reactions per laser shot, just close to the level of experimental sensitivity,  $10^6$   $p\text{-}^{11}\text{B}$  fusion reactions,  $10^9$  fusion reactions/sr and  $10^{10}$  fusion reactions/sr, respectively. The fusion gains of [20, 21], from laser pulsed-driven fusion experiments (500 J, 0.3 ns), at the Prague Asterix Laser System (PALS), without any special conditions for the preparation of pulse contrast, suppression of pre-pulses, etc, were too high compared to preceding knowledge and led to the estimation of their way of achievement [23, 24].

In the theoretical works, of [23–26], it is stated that the increase from  $10^5$  to  $10^{11}$  experimentally produced alphas [18–21] is due to the rise of the  $p\text{-}^{11}\text{B}$  fusion species temperature (chain reaction alpha heating effect), to temperatures corresponding to the optimum  $p\text{-}^{11}\text{B}$  cross section (close to 675 keV resonance) and the manifestation of the avalanche effect. This interpretation was followed by the criticism of [27] and the consequent answer of [28] on the subject, as well as by the paper publication of [25]. The latter enabled the proposal of an alternative avalanche model (new evaluation on the avalanche time), at the low fusion cross section resonance of 163 keV. The main conclusion of the aforementioned interpretation cases is that the relation between the important increase of the fusion produced alphas and the chain reactions alpha heating effect and the related avalanche effect, remains as a main controversial issue. According to [14, 15, 29], the avalanche effect is manifested, when the produced alpha particle density is  $\sim 2$  orders of magnitude lower than the initial  $p\text{-}^{11}\text{B}$  medium density. A natural equivalent macroscopic example of the alpha particles avalanche is the snowball avalanche, where a small snowball coming down the hillside, increases its amount of snow and finally results in the well-known avalanche phenomenon (figure in transparency #3 from N. Nissim) [30]. The avalanche term is used for both non-LTE and LTE cases, where the macroscopic effect is described by an alpha multiplication factor  $N(t > 0)/N(t = 0) = \exp \int_0^t \zeta dt$  [31]. Thus, in the context of the present work, the expression “chain reactions alpha heating effect and the related avalanche effect” is used as an appropriate and explicit term, for the description of the energy transfer from the three (3) fusion produced alphas (of 2.9 MeVs initial energy, each) to the  $p\text{-}^{11}\text{B}$  fusion medium species. This energy transfer results to the improvement of the fusion reaction rate (RR) and the fusion gain, as well

as to the approximation of break-even ( $Q = P_{\text{fus}}/P_{\text{Brems}} = 1$ ) and self-sustained ignition conditions ( $Q = P_{\text{fus}}/P_{\text{Brems}} > 1$ ), under suitable p- $^{11}\text{B}$  medium initial conditions of density and temperature.

In the context of the present work, the study of the chain reactions alpha heating effect doesn't include the introduction of secondary (intermediate) nuclear reactions between the p- $^{11}\text{B}$  fusion born alphas and the  $^{11}\text{B}$  atoms of the medium, as in [32], but only elastic binary collisions between the alphas and the p,  $^{11}\text{B}$  fusion species, as initially proposed in [23–25]. The study of the chain reactions plasma heating effect in a low-density ( $n \sim 10^{19} \text{ m}^{-3}$ – $10^{20} \text{ m}^{-3}$ ) p- $^{11}\text{B}$  medium is of particular interest, as the last few years, there are a lot of private companies and universities, devoting extensive effort on the manufacturing of Compact Magnetic Fusion Devices (CMFDs), operating with plasma densities in the range  $10^{19} \text{ m}^{-3}$ – $10^{21} \text{ m}^{-3}$ . Compact Magnetic Fusion Devices present: i) A small operational volume, ii) A lower cost for the device and operation, compared to the currently existing magnetic and inertial confinement fusion devices and iii) Allow the experimental investigation of different fusion fuels and the alpha plasma heating effect [33]. Typical examples of the developing Compact Magnetic Fusion Devices belong to General Fusion [34, 35], Helion Energy [36], Tri-Alpha Energy [37] and Lockheed Martin [38], as well as to the ENN University in China (Helong experiment) [39]. In the aforementioned Compact Magnetic Fusion Devices, the magnetic trapping of the plasma is ensured, applying mirror-like magnetic configurations in combination with the more complex Field Reversed Configuration (FRC) [40]. Compact Magnetic Fusion Devices may also use externally injected negative-neutral beams of protons [37], for the sustainment of the medium at high temperature, for the continuation of the fusion reactions in the device. Even though the majority of the Compact Magnetic Fusion Devices is based on a greater extent on D-T and D- $^3\text{He}$  fusion, valuable information concerning their operation regime and their plasma energy confinement time ( $\tau_E$ ), can be extracted and applied for the numerical study of the chain reactions alpha heating effect in a relative Compact Magnetic Fusion Device with low plasma medium density ( $n \sim 10^{19} \text{ m}^{-3}$ – $10^{21} \text{ m}^{-3}$ ). Although the numerical simulations of the present work don't refer to a specific potential experimental setup or device, they may refer to these proposed magnetic trapping schemes, where physical processes, such as the chain reactions effect and the related avalanche effect may occur.

The aim of the present numerical study is to show the temporal evolution of the different physical parameters and the appearance of the physical processes in a low-density p- $^{11}\text{B}$  medium ( $n = 10^{20} \text{ m}^{-3}$ – $10^{21} \text{ m}^{-3}$ ), with specifically selected boron to proton density ratios,  $n_p/n_B > 1$ , and initial temperature in the range of 80 keV–350 keV. According to [4, 10, 14, 41], density ratios are favorable for the optimization of Bremsstrahlung radiation losses, in a wide range of initial p- $^{11}\text{B}$  medium densities {e.g. low ( $n \sim 10^{19}$ – $10^{20} \text{ m}^{-3}$ ), intermediate ( $n \sim 10^{25} \text{ m}^{-3}$ ) or solid-state}. Thus, the selection of the aforementioned p- $^{11}\text{B}$  medium initial conditions of density and temperature was based on the criteria of: i) Bremsstrahlung losses reduction and ii) The time interval, in which the alpha particle density becomes significant for the achievement of  $Q = P_{\text{fus}}/P_{\text{Brems}} > 1$ , due to the manifestation of the avalanche effect, to be comparable with the typical time of medium expansion. For the optimization of the two above criteria and the achievement of break-even and self-sustained fusion ignition conditions, we select to work with initial low-density p- $^{11}\text{B}$  plasma ( $n_p = 1.0 \times 10^{20} \text{ m}^{-3}$ ) with the optimum density ratio  $n_p/n_B$  of figure 1 and initial temperature in the range between 150 keV–350 keV. For the case of an initial p- $^{11}\text{B}$  medium temperature, lower than  $T_{\text{in}} = T_P = T_B = T_e = 100 \text{ keV}$ , according to figure 2, there is no ignition, due to the continuous decrease of the fusion species temperature, resulting from the relatively low alpha density production.

## 2 Numerical investigation of the chain reactions alpha heating effect in a low-density p-<sup>11</sup>B medium

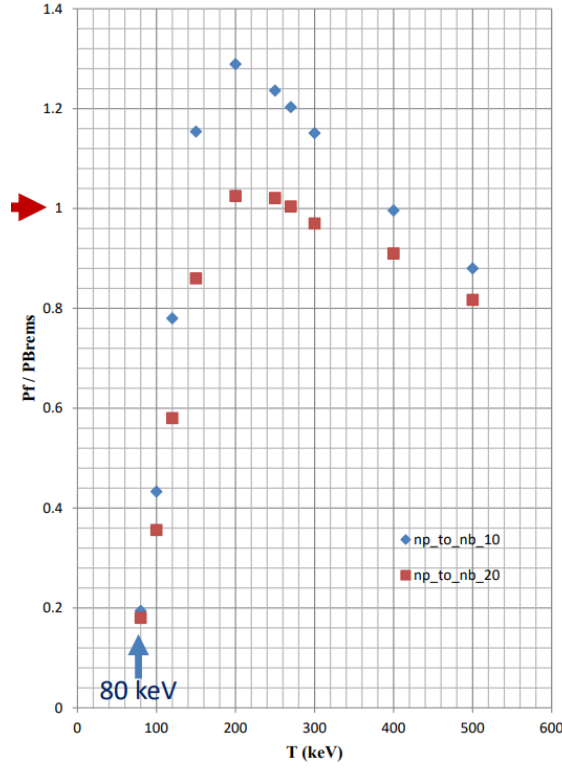
In [42–44], the main equations of a multi-fluid, global particle and energy balance code, which can be applied in both magnetically and inertially confined fusion plasmas, are presented (see appendix A and B). The described mathematical model: i) Includes collisions between the medium species, ii) Treats reacting ions, electrons and fusion born alphas, as four (4) separate fluids and iii) Calculates the temporal evolution of the physical parameters of the fusion medium (fusion reactivity, fusion reaction rate, species densities and internal energies), for different initial conditions of density and temperature of the fusion medium and (iv) include Bremsstrahlung radiation losses. The collision frequencies, for the description of collisions between the medium species (reactant ions, alphas and electrons), employ the formulations of [45], that are confirmed by [46]. The latter reference examines up to twenty (20) references, for plasma species in local thermodynamic equilibrium (LTE). As it was mentioned in previous paragraphs of the introduction, in the case of the p-<sup>11</sup>B nuclear fusion reaction, the chain reactions alpha heating effect concerns the collisional energy transfer from the three (3) fusion produced alphas to the fusion species (p, <sup>11</sup>B) and the consequent important rapid rise of the temperature of the latter, to temperatures corresponding to the optimum p-<sup>11</sup>B fusion cross section. The temperature increase of the fusion p, <sup>11</sup>B species results to the important increase of the alpha density production and the consequent significant enhancement of the  $Q = P_{\text{fus}}/P_{\text{Brems}}$  ratio. Thus, the inclusion of collisions and Bremsstrahlung losses in the dynamical, multi-fluid global particle code is a valuable tool for the examination and the identification of the chain reactions alpha heating effect and the related avalanche effect in a p-<sup>11</sup>B medium. In the context of the model, the p-<sup>11</sup>B fusion reactivity calculations exploit the cross section parametric equation of [11] with the improved and/or updated values of [47]. The interest in the use of the updated cross section values of [47] is the optimization of the Qratio, as presented in [10].

As it was already mentioned in the Introduction section, the major difficulty in the acceptance of p-<sup>11</sup>B nuclear fusion reaction as a clean energy production source, is its extremely high required fusion ignition temperature, in the range between 350 keV and 600 keV, compared to the low fusion ignition energy of D-T nuclear fusion reaction 30 keV–50 keV. The energy gap that is pointed out between these two fusion ignition energies is actually the reason that the only p-<sup>11</sup>B experimental facilities that have been developed so far, concern the interaction of an energetic proton beam with a solid <sup>11</sup>B target (or plasma) and not the study of p-<sup>11</sup>B fusion oriented experiments, as in D-T fusion (ITER, NIF). However, the numerical simulations of the present work show that, if the p-<sup>11</sup>B has the optimum proton to boron density ratio of figure 1 and the fusion process is initiated with temperatures in the range of 150 keV–350 keV ( $T_{\text{in}} > 100$  keV) (low cross-sections or reactivities in this initial temperature range), the chain reactions alpha heating is going to lead gradually to the rise of the p, <sup>11</sup>B fusion species temperature, and consequently to the improvement of the fusion reactivity and the increase of the alpha density production.

## 3 Numerical simulation results

In figure 1, the value of the ignition criterion ( $Q = P_{\text{fus}}/P_{\text{Brems}}$ ), as a function of the p-<sup>11</sup>B fusion medium initial temperature, is presented, for two (2) different proton to boron density ratios ( $\varepsilon = n_p/n_B$ ):  $n_p/n_B = 10$  (blue points) and  $n_p/n_B = 20$  (red points). The initial proton density of

$n_p = 10^{20} \text{ m}^{-3}$  is similar to the plasma density in Compact Magnetic Fusion Devices (CMFDs) or Tokamaks. Our numerical results with the use of the multi-fluid code and the consideration of the latter initial ion density, show that the  $Q$  value is higher than 1 ( $Q > 1$ ), for temperatures in the range of 150 keV–300 keV and lower than 1 ( $Q < 1$ ) for higher temperatures. For both cases of  $n_p/n_B$  (10 and 20), the ignition criterion value,  $Q$ , is maximized at the temperature of 200 keV. However, in the case of  $n_p/n_B = 10$ , the  $Q$  value is higher ( $Q \sim 1.3$ ). Therefore, for the presentation and the description of our numerical simulations, the density ratio of  $n_p/n_B = 10$  is selected.

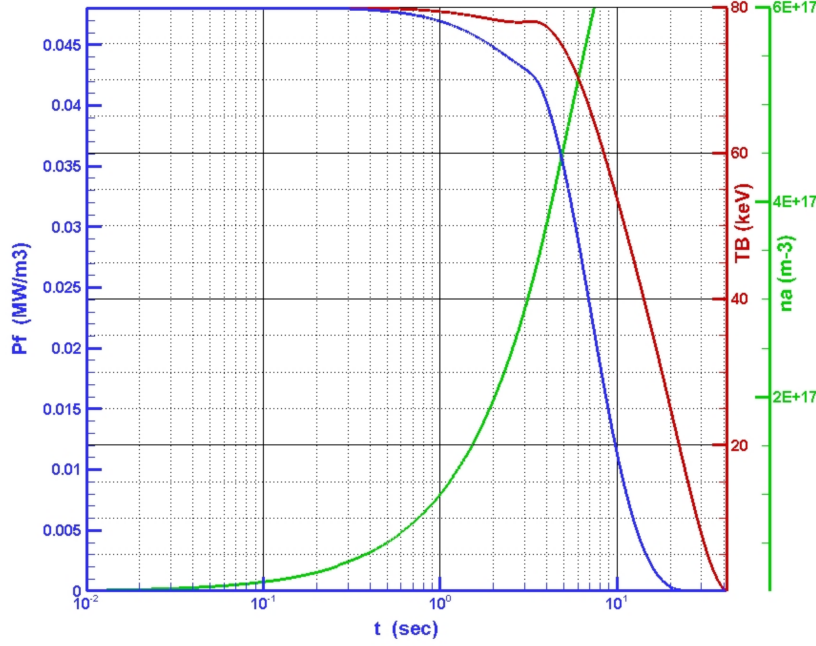


**Figure 1.** Ignition criterion optimization ( $Q = P_{\text{fus}}/P_{\text{Brems}}$ ) in low-density p- $^{11}\text{B}$  fuel ( $n_p = 1.0 \times 10^{20} \text{ m}^{-3}$ ), through the consideration of different density ratios ( $\varepsilon = n_p/n_B$ ) between the p,  $^{11}\text{B}$  fusion species. The numerical results show that the optimum density ratio, which leads to the highest values of the ignition criterion (self-sustained fusion ignition conditions:  $Q > 1$ ) is the  $n_p/n_B = 10$ .

### 3.1 Neutral low density p- $^{11}\text{B}$ medium with $T_{\text{in}} = T_p = T_B = T_e = 80 \text{ keV}$ and $n_p/n_B = 10$

We initiate our numerical study with the case of the initial p- $^{11}\text{B}$  medium temperature of  $T_{\text{in}} = T_p = T_B = T_e = 80 \text{ keV}$ , which corresponds to a low value of the ignition criterion, ( $Q \sim 0.2$ ), as shown in figure 1. The relevant numerical results, concerning the temporal evolution of the fusion power density ( $P_f$ ), the alpha density production ( $n_\alpha$ ) and the  $^{11}\text{B}$  Boron temperature ( $T_B$ ), are presented in figure 2. The main important result of this figure is that, the initial  $^{11}\text{B}$  temperature of  $T_{\text{in}} = 80 \text{ keV}$  decreases slightly, until approximately 10 sec. After this time interval, the  $^{11}\text{B}$  temperature (red curve) decreases very fast to reach zero, due to Bremsstrahlung radiation losses. In the time interval of 10 sec, the produced alpha density (green curve) is not high enough, in order to transfer sufficient amounts of energy to the p,  $^{11}\text{B}$  fusion species and thus increase their temperature, to values corresponding to the





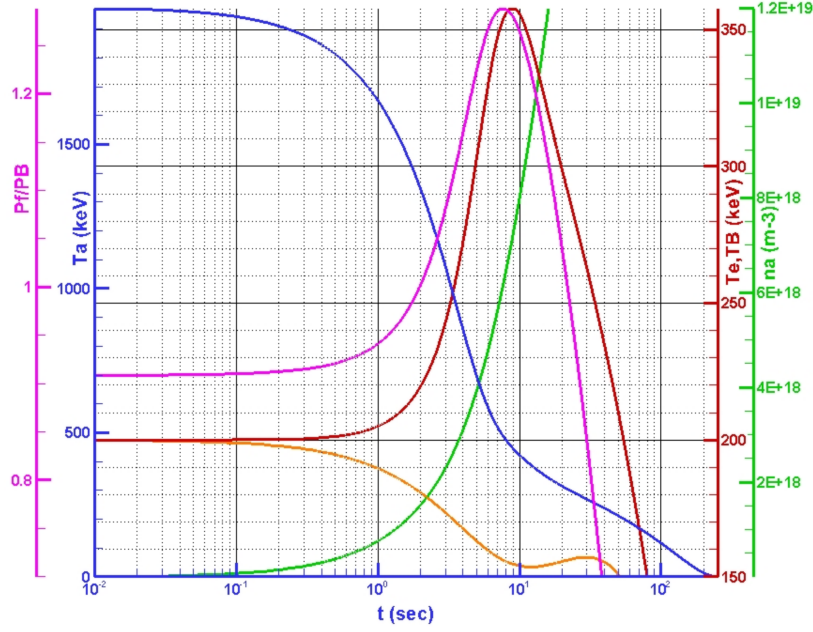
**Figure 2.** Temporal evolution of Fusion Power Density (blue curve), alpha density (green curve) and Boron temperature (red curve), for  $n_p = 1.0 \times 10^{20} \text{ m}^{-3}$ ,  $n_p/n_B = 10$  and  $T_{\text{in}} = 80 \text{ keV}$ . Continuous decrease of the fusion species temperature, no ignition, absence of the chain reactions alpha heating effect.

optimum p- $^{11}\text{B}$  fusion cross section (absence of the avalanche alpha heating effect). Therefore, in the case of  $T_{\text{in}} = 80 \text{ keV}$  initial medium temperature, there is no fusion ignition and burning of the fuel.

Figure 3 shows the temporal evolution of the medium (p,  $^{11}\text{B}$ ) species temperature for the initial conditions: i) Proton density  $n_p = 10^{20} \text{ m}^{-3}$ , ii)  $n_p/n_B = 10$  (optimum density ratio of figure 1) and iii)  $T_{\text{in}} = T_p = T_B = T_e = 200 \text{ keV}$ . The first important observation is that at  $10^{-1} \text{ sec}$ , there is fast Boron temperature increase, from  $T_B = 200 \text{ keV}$  to  $T_B = 350 \text{ keV}$ , which lasts until  $t \sim 10 \text{ sec}$ . The increase in the Boron temperature occurs at a characteristic time, for which, the produced alpha particle density,  $n_\alpha$ , is approximately  $\sim 2$  orders of magnitude lower than the initial Boron density. For the time interval ( $\Delta t$ ) between  $10^{-1} \text{ sec}$  and  $10 \text{ sec}$ , the produced alpha particle density increases from  $\sim 10^{18} \text{ m}^{-3}$  to  $\sim 1.2 \cdot 10^{19} \text{ m}^{-3}$ , which is very close to the initial boron density. At the time point of  $10^{-1} \text{ sec}$ , the rapid increase in the  $^{11}\text{B}$  temperature and the alpha density production are the consequences of the chain reactions alpha heating effect and the related avalanche effect. The manifestation of the avalanche effect is related with the specific value of the produced alpha density (2 orders of magnitude lower than the initial Boron density) and the beginning of the important rise in the p,  $^{11}\text{B}$  fusion species temperature. From  $10^{-1} \text{ sec}$  to  $10 \text{ sec}$ : i) The important rise of the fusion species temperature enables the continuous increase of the ignition criterion value,  $Q = P_{\text{fus}}/P_{\text{Brems}}$ , until its maximization at  $10 \text{ sec}$  ( $Q \sim 1.3$ ). ii) The alpha temperature decreases, as a result of the alpha energy transfer to the p,  $^{11}\text{B}$  fusion species and the electrons. This specific correlation, concerning the alpha energy transfer, is a significant indicator of the chain reactions alpha heating effect.

The second important observation arising from figure 3, concerns the decoupling between the electron temperature ( $T_e$ ) and the  $^{11}\text{B}$  Boron species temperature ( $T_B$ ), which begins at  $\sim 10^{-1} \text{ sec}$ . The decrease in the electron temperature starts to occur, when: i) The Boron temperature begins to rise,



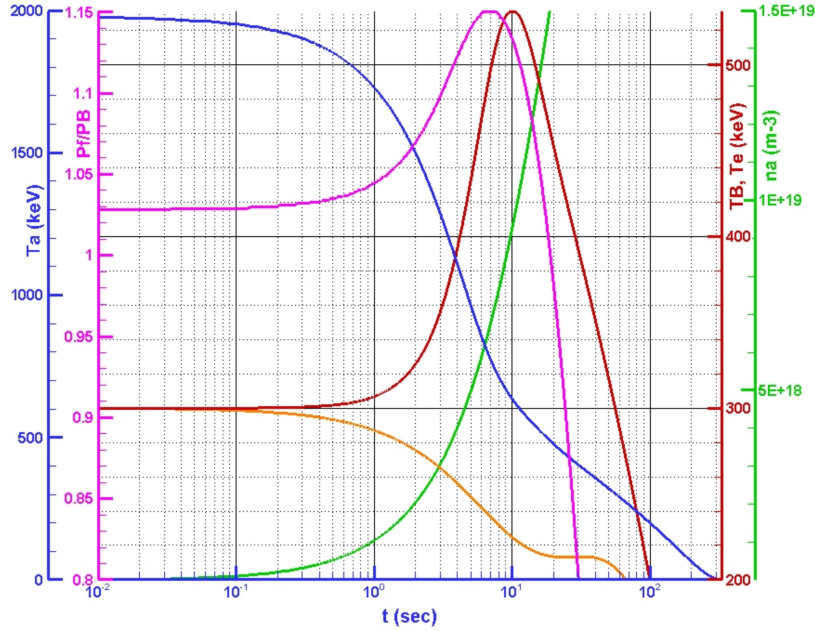


**Figure 3.** Temporal evolution of  $Q = P_{\text{fus}}/P_{\text{Brems}}$  (pink curve), alpha density (green curve), Boron temperature (red curve) and electron temperature (orange curve), for  $n_p = 1.0 \times 10^{20} \text{ m}^{-3}$ ,  $n_p/n_B = 10$  and  $T_{\text{in}} = 200 \text{ keV}$ . Rapid and significant improvement in the  $^{11}\text{B}$  species temperature, the alpha density and the value of the ignition criterion ( $Q = P_{\text{fus}}/P_{\text{Brems}}$ ), between  $t = 10^{-1} \text{ sec}$  and  $t \sim 10 \text{ sec}$ , due to the manifestation of the avalanche effect. In the same time interval, self-sustained ( $Q > 1$ ) fusion ignition conditions are obtained. At the maximum value of the  $Q = P_{\text{fus}}/P_{\text{Brems}}$  ( $Q \sim 1.3$ ),  $P_{\text{fus}} = 0.35 \text{ MW/m}^3$ .

ii) The produced alpha particle density is about ( $\sim$ ) 2 orders of magnitude lower than the initial boron density and iii) The alpha energy starts to decrease. Consequently, the characteristic time of  $10^{-1} \text{ sec}$  relates the Bremsstrahlung radiation losses (decrease of  $T_e$ ) with the energy transfer from the alphas to the electrons and the p,  $^{11}\text{B}$  fusion species, indicating once again, the existence of the chain reactions alpha heating effect. The electrons lose energy (Bremsstrahlung losses), but they also (in parallel) absorb energy from the fusion species via collisions. The expected result would be the decrease of the fusion species (p,  $^{11}\text{B}$ ) temperature. However, this energy loss is replenished by the energy transfer from the alphas. Therefore, there is a strong correlation, between the energy transfer from the alphas to the fusion species and the electrons. The majority of the alpha energy is transferred to the fusion species (p,  $^{11}\text{B}$ ), resulting in their fast temperature increase to values ( $T \sim 400 \text{ keV}$ ), which are near to the optimum p- $^{11}\text{B}$  fusion cross-section (figure 3). The temperature gap between the maximum Boron temperature ( $T_B$ ) and the minimum electron temperature ( $T_e$ ) varies in time. The maximum value of  $T_B/T_e > 2$  is observed at  $t \sim 10 \text{ sec}$ , where a self-sustained fusion with  $Q \sim 1.3$  is achieved. In other works [4, 10, 13], the  $T_e < T_B$  relation is given as an initial condition and doesn't arise from self-consistent calculations. In contrast, in our simulations, the significant temperature gap of  $T_e < T_B$  results from the use of the self-consistent multi-fluid global particle and energy balance code (with collisions between all the species and Bremsstrahlung losses). This important self-consistent numerical result confirms that the enhancement of the fusion power density ( $P_{\text{fus}}$ ) with  $Q > 1$  (fusion ignition and self-sustained process) is strongly related with the relation:  $T_B > 2T_e$ . The fusion power density ( $P_{\text{fus}}$ ), corresponding to the maximum p,  $^{11}\text{B}$  fusion species

temperature, is  $P_{\text{fus}} \sim 0.35 \text{ MW/m}^3$ . If we compare the fusion power densities of  $T_{\text{in}} = 80 \text{ keV}$  (no fusion ignition) and  $T_{\text{in}} = 200 \text{ keV}$  (ignition), we obtain an improvement factor of 100, with only 2.5 times increase in the initial p- $^{11}\text{B}$  medium temperature.

The initial p- $^{11}\text{B}$  medium temperature of  $T_{\text{in}} = 300 \text{ keV}$  (figure 4) corresponds to the right limit of the temperatures range of figure 1, for which there is ignition and self-sustainment ( $Q = P_{\text{fus}}/P_{\text{Brems}} > 1$ ) of the low-density fusion medium. Figure 4 shows the temporal evolution of the physical parameters of the p- $^{11}\text{B}$  medium (alpha temperature, alpha density,  $^{11}\text{B}$  temperature, electron temperature, ignition criterion value), for proton density:  $n_p = 10^{20} \text{ m}^{-3}$  and proton to boron density ratio:  $n_p/n_B = 10$ . The physical effects, concerning: i) The alpha energy transfer to the p,  $^{11}\text{B}$  fusion species (chain reactions alpha heating effect), ii) The enhancement of the ignition criterion,  $Q = P_{\text{fus}}/P_{\text{Brems}}$ , iii) The decrease in the electron temperature ( $T_e$ ), iv) The temperature gap between the p,  $^{11}\text{B}$  fusion species and the electrons ( $T_B/T_e > 2$ ), occurs in a characteristic time interval of  $\Delta t \sim 10 \text{ sec}$ , which is similar to that of the case of  $T_{\text{in}} = 200 \text{ keV}$  initial p- $^{11}\text{B}$  medium temperature.



**Figure 4.** Temporal evolution of  $Q = P_{\text{fus}}/P_{\text{Brems}}$  (pink curve), alpha density (green curve), Boron temperature (red curve) and electron temperature (orange curve), for  $n_p = 1.0 \times 10^{20} \text{ m}^{-3}$ ,  $n_p/n_B = 10$  and  $T_{\text{in}} = 300 \text{ keV}$ . Similar results with the case of  $T_{\text{in}} = 200 \text{ keV}$  are observed, but with the differences that the  $^{11}\text{B}$  species temperature reaches higher values in the same interval  $\Delta t$  ( $10^{-1} \text{ sec}$  to  $10 \text{ sec}$ ), due to the chain reactions alpha heating effect. The fusion process is self-sustained ( $Q > 1$ ), from the beginning of the process and until  $t \sim 10 \text{ sec}$ . At the maximum value of the ignition criterion,  $Q = P_{\text{fus}}/P_{\text{Brems}}$ ,  $P_{\text{fus}} = 0.6 \text{ MW/m}^3$ .

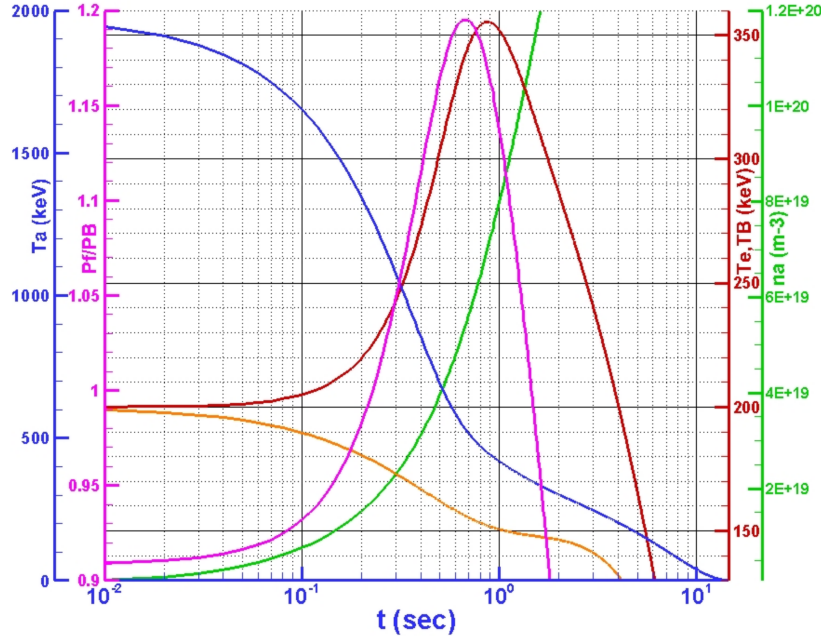
However, the main differences of figure 3 ( $n_p = 10^{20} \text{ m}^{-3}$ ,  $n_p/n_B = 10$ ,  $T_{\text{in}} = 200 \text{ keV}$ ) and figure 4 ( $n_p = 10^{20} \text{ m}^{-3}$ ,  $n_p/n_B = 10$ ,  $T_{\text{in}} = 300 \text{ keV}$ ) are: i) The value of the ignition criterion,  $Q = P_{\text{fus}}/P_{\text{Brems}}$ , which is higher than 1 from the beginning of the fusion process in the latter case, ii) The Boron maximum temperature,  $T_B \sim 540 \text{ keV}$ , is 1.4 times higher than that of  $T_{\text{in}} = 200 \text{ keV}$ . On the other hand, the alpha temperature and the alpha density values, corresponding to the maximum Boron temperature ( $^{11}\text{B}$ ), are very close in both cases. In the case of  $T_{\text{in}} = 300 \text{ keV}$ , at the p,  $^{11}\text{B}$  fusion species temperature maximization, the produced fusion power density is  $P_{\text{fus}} \sim 0.6 \text{ MW/m}^3$ .

The latter value is approximately two (2) times higher than that of  $T_{\text{in}} = 200$  keV, due to the higher p,  $^{11}\text{B}$  fusion species temperatures (thus, higher p- $^{11}\text{B}$  fusion cross section). However, in the case of  $T_{\text{in}} = 300$  keV, the interest in producing a higher amount of fusion power density ( $P_{\text{fus}}$ ) is reduced by the fact that, the initial medium temperature has to be increased by 100 keV. This means that more power for the auxiliary systems of the fusion device may be needed.

### 3.2 Potential improvements of the Fusion Power Density ( $P_{\text{fus}}$ )

The p- $^{11}\text{B}$  medium density increase to  $n_p = 10^{21} \text{ m}^{-3}$  ( $n_p/n_B = 10$ ) for plasma formation in a Compact Magnetic Fusion Device, constitutes an important and hard challenge to be achieved. Nonetheless, it presents major interest for a higher fusion power density production ( $P_{\text{fus}}$ ), aiming to energy generation applications. Figure 5 depicts the temporal evolution of the species temperatures (alpha,  $^{11}\text{B}$  Boron and electrons) and the ignition criterion ( $Q = P_{\text{fus}}/P_{\text{Brems}}$ ), for the case of  $n_p = 10^{21} \text{ m}^{-3}$ ,  $n_p/n_B = 10$  and  $T_{\text{in}} = 200$  keV. The description of the temporal evolution of the physical processes, concerning: i) The chain reactions alpha heating effect, ii) The rise of the p,  $^{11}\text{B}$  fusion species temperatures, iii) The improvement of the  $P_{\text{fus}}/P_{\text{Brems}}$  ratio, iv) The temporal decoupling between the electron and the  $^{11}\text{B}$  ion temperatures, v) The temperature gap between the ion and the electron temperatures ( $T_B/T_e > 2$ ), vi) The self-sustained fusion ignition conditions and vii) The manifestation of the avalanche effect, remain similar with the aforementioned cases of section 3.1 (Initial density:  $n_p = 10^{20} \text{ m}^{-3}$ ,  $n_p/n_B = 10$  and Initial medium temperature:  $T_{\text{in}} = 200$  keV–300 keV). However, the main difference between these two (2) initial medium density cases concerns the characteristic time, that the main physical processes appear. The latter is ten (10) times shorter:  $t = 0.65 \text{ sec} < 1 \text{ sec}$ , in the case of  $n_p = 10^{21} \text{ m}^{-3}$  (figure 5). Also, in the case of  $n_p = 10^{21} \text{ m}^{-3}$ , ( $n_p/n_B$ ) = 10 and  $T_{\text{in}} = 200$  keV, the maximum Boron temperature:  $T_B \sim 360$  keV, corresponds to:  $Q = P_{\text{fus}}/P_{\text{Brems}} \sim 1.2$  and  $P_{\text{fus}} \sim 40 \text{ MW/m}^3$ . This latter value is  $10^2$  times higher, compared to that of figure 3 ( $n_p = 10^{20} \text{ m}^{-3}$ , ( $n_p/n_B$ ) = 10 and  $T_{\text{in}} = 200$  keV), due to the higher corresponding alpha density production ( $n_\alpha = 1.2 \cdot 10^{20} \text{ m}^{-3}$ , in relation with  $n_\alpha = 1.2 \cdot 10^{19} \text{ m}^{-3}$  of figure 3).

The  $Q = P_{\text{fus}}/P_{\text{Brems}}$  ratio could be improved, employing recovery techniques, already proposed for Compact Fusion Devices or Tokamaks. More specifically, the radiation energy loss could be converted into electricity, through the use of an appropriate configuration, as that proposed for the case of the plasma focus device [48]. In this configuration, an external specifically shaped blanket of materials, covering the device, allows the conversion of radiation into electrical energy, with a higher than 50% efficiency [48]. The recovery energy could drive an auxiliary power source, for the heating of the medium species (e.g. electrons or ions) and the improvement of the output power gain of the fusion device [4]. For a compact magnetic fusion configuration, the covering of the device with an appropriate blanket is more feasible, compared to the case of huge devices (e.g. Tokamaks). Tokamak machines, for example, use external auxiliary microwave devices for the heating of the plasma electrons or ions. Other auxiliary sources for plasma heating include the injection of high energy neutral beams [37]. The potential development and use of auxiliary sources, the operational energy of which will come from recovery energy losses, would be an important progress for the development of high fusion gain compact devices. Indicatively, we mention that, if for the case of  $n_p = 10^{20} \text{ m}^{-3}$ , ( $n_p/n_B$ ) = 10 and  $T_{\text{in}} = 200$  keV, a 50% potential recovery coefficient is assumed, the improved ignition criterion value would be:  $Q = P_{\text{fus}}/P_{\text{Brems}} \sim 3$ , with  $P_{\text{fus}} \sim 0.57 \text{ MW/m}^3$ .



**Figure 5.** Temporal evolution of  $Q = P_{\text{fus}}/P_{\text{Brems}}$  (pink curve), alpha density (green curve), Boron temperature (red curve) and electron temperature (orange curve), for  $n_p = 1.0 \times 10^{21} \text{ m}^{-3}$ ,  $n_p/n_B = 10$  and  $T_{\text{in}} = 200 \text{ keV}$ . The one (1) order of magnitude increase in the initial medium density results in self-sustained fusion ignition conditions in  $\Delta t \sim 0.2 \text{ sec}$ . At  $t \sim 0.65 \text{ sec}$  ( $Q \sim 1.2$ ), the fusion power density production corresponds to  $P_{\text{fus}} = 40 \text{ MW/m}^3$ .

#### 4 Discussion-conclusions

Our numerical simulation results, with the use of a multi-fluid code, for initial (p,  $^{11}\text{B}$ ) fusion medium density:  $n_p = 10^{20} \text{ m}^{-3}$  (or  $10^{21} \text{ m}^{-3}$ ) and the consideration of the optimum density ratio of figure 1:  $(n_p/n_B) = 10$ , enable the determination of the initial temperature range:  $150 \text{ keV} \leq T_{\text{in}} \leq 350 \text{ keV}$ , in which the  $Q = (P_{\text{fus}}/P_{\text{Brems}})$  ratio is higher than 1 ( $Q > 1$ : self-sustained ignition conditions of the p- $^{11}\text{B}$  medium).

The numerical results of the code concern only temporal evolution of the physical parameters, without plasma expansion (or losses) during the fusion process. This is an acceptable and good approximation for the case of Compact Magnetic Fusion Devices or Tokamaks. These two types of devices combine magnetic fields for the formation of toroidal “magnetic bottles”, confining the hot and low density plasma in a stable way for long time intervals. The stable alpha production, during the first p- $^{11}\text{B}$  fusion oriented experiment by TAE [50], using a compact Stellarator coupled with an energetic proton beam, confirms the efficient plasma trapping without losses, for relatively long periods of time [37]. Open configurations, such as the mirror-like device, suffer from plasma losses, but they are cost effective and simple in operation for near future p- $^{11}\text{B}$  fusion oriented experiments in laser facilities. However in some confinement scenarios (e.g mirror-like magnetic devices), concerning low density plasma with negligible radiation re-absorption and anisotropic distribution function for electrons, the Bremsstrahlung radiation can be substantially reduced [51]. In the present work, we include only fluids described by Maxwellian distribution.

The multi-fluid model (see appendix A and B) does not include electromagnetic (Bremsstrahlung radiation) interaction with the fusion medium electrons, which could thus transfer energy by re-

absorption. The Bremsstrahlung radiation is only coupled with the electron fluid (see appendix A and B) and refers to radiation losses. The initial p-<sup>11</sup>B medium density, used in the code, is  $10^{20} \text{ m}^{-3}$  or  $10^{21} \text{ m}^{-3}$ , which is close to the density of Compact Magnetic Fusion Devices or Tokamaks. In this density range, the radiation emission corresponds to a broadband spectrum of x-rays (200 eV–10 keV), including hard x-rays (20 keV–200 keV: Bremsstrahlung emission), due to supra-thermal electrons. The last emission is used as a diagnostic to probe the plasma conditions (Tomographic measurement). For a typical p-<sup>11</sup>B medium temperature in the range of 20 keV–200 keV, the wavelength for the maximum emitted radiation ( $\lambda_{\text{max}}$ ), corresponds to 0.03 nm–0.003 nm [52]. The majority of the emitted radiation energy occurs in the immediate spectral region of the  $\lambda_{\text{max}}$  wavelength, while for longer wavelengths, it drops by at least two orders of magnitude. If we assume a length,  $x \sim 10 \text{ m}$ – $15 \text{ m}$ , for the fusion medium (compact device) and consider the expression (34) of [53] for the Bremsstrahlung absorption coefficient,  $\kappa$ , (and/or optical depth), the relation  $I = I_0 \exp(-\kappa x)$  gives very small radiation absorption.

However, if the fusion medium initial density increases to solid density or higher (compression by 50–100 times), the re-absorption of Bremsstrahlung radiation is not negligible and must be considered. In this case, the energy transfer from Bremsstrahlung radiation to electrons will modify the collision frequency  $\nu_{ei}$ , with potential improvement of the  $P_{\text{fus}}/P_{\text{Brems}}$  ratio. Two different theoretical schemes for fusion with solid state p-<sup>11</sup>B medium density are proposed in the international literature. The first concerns the scheme of HB11 Energy [49] and the second, a hybrid-like configuration [13, 54]. In the two aforementioned schemes, the high density plasma expands very fast in a few tens of ps and the manifestation of the effects presented in this work, especially the chain reactions alpha heating effect and the related avalanche effect occur in relatively longer time intervals. For this reason, the HB11 Energy scheme [49] proposes the application of a laser produced high magnetic field (500 T or higher) for at least 10 ns, for plasma trapping. In the relative case of the hybrid-like scheme, the compression of the solid density target by at least 50 times is suggested [13, 15]. Under these initial conditions, an important contribution of the chain reactions alpha heating effect and the related avalanche effect is expected, that will in turn allow the increase of the fusion species temperatures to values corresponding to the optimum p-<sup>11</sup>B reaction cross section and the fusion ignition in the fuel. For both schemes, the re-absorption of a part of the emitted Bremsstrahlung radiation necessitates the re-evaluation of the collision frequency expressions. However, a multi-fluid model may at least evaluate the necessary time for the trapping and the manifestation of the important alpha heating effects.

In the low density case (e.g. Tokamak machines or compact devices), a deviation from the Maxwellian distribution may appear, during processes generating energetic species, which are not in equilibrium with the background plasma. Hot electrons (and/or accelerated ions), penetrate in the background fluid and lose energy through Coulomb interactions. For electrons, a deviation from the Maxwellian distribution is related for example, with the presence of gradients of density and temperature, in open configurations and near the divertor wall of the Tokamak machine. Energetic electrons are less collisional and could transfer most of the heat, thus producing an asymmetrical Maxwellian distribution. In the same context, another issue of the thermal equilibrium and the multi-fluid model, concerns the generation of energetic alphas in the fusion medium, due to nuclear reactions. We assume the approximation that the alphas are described as a separate fluid with Maxwellian distribution.

For the evaluation of the temporal evolution of the alpha fluid internal energy density, in the multi-fluid model (see appendix A and B), the total alpha energy of 8.7 MeV is used, assuming iso-energetic alpha production, with the cross section formulation, corresponding to the improved values from [47].



For the production of the three alphas from the  $p\text{-}^{11}\text{B}$  fusion reaction, an alternative nuclear process (channels) has been presented, in relation to the one producing a direct breakup with three iso-energetic (2.9 MeV) alphas with 8.7 MeV total energy. In this alternative process [8, 12, 55, 56], the  $p\text{-}^{11}\text{B}$  nuclear reaction leads initially to the formation of  $^{12}\text{C}^*$ . The latter breakups into an alpha particle plus a  $^8\text{Be}$  in the ground or the first excited state, which then decays to alpha particles with different energy spectra. The total energy of the three produced alphas is also 8.7 MeV and the energy spectrum of the produced alphas is continuum, with a maximum value of approximately 6 MeV [8, 12, 56]. This fact could be a good argument for the interpretation of the experimental results on alpha particle production of [19, 57] and [20, 21] at PALS laser facility, where the measured alpha energy achieved values of 4.5 MeV–5 MeV. However the measured alpha spectra, varies between the aforementioned experiments, due to the different experimental conditions and setups, so there is no a clear conclusion. Secondary nuclear reactions [8, 19, 58, 59], produce undesirable neutrons or gamma rays for fusion of  $p\text{-}^{11}\text{B}$ , but with low efficiency.

The temporal correlation between the increase of the  $p, ^{11}\text{B}$  fusion species temperature and the decrease of the alpha temperature is an important numerical result, as it allows the establishment of the chain reactions alpha heating effect and the related manifestation of the avalanche effect. The temperature rise of the  $p, ^{11}\text{B}$  fusion species results in the density increase of the produced alphas, due to the higher corresponding value nuclear reaction cross-section. This means that more and more alpha particles are produced, increasing the amount of energy transfer to the  $p, ^{11}\text{B}$  fusion species. The manifestation of the avalanche effect is strongly related with a critical value of the alpha density production. More specifically, the avalanche effect occurs, when the produced alpha particle density is  $\sim 2$  orders of magnitude lower than the initial Boron density. The energy transfer from the alphas to the  $p, ^{11}\text{B}$  fusion species results in (i) The replenishment of the energy that the latter ( $p, ^{11}\text{B}$ ) lose, due to their collisions with the electrons and (ii) The fast increase of their temperatures, to temperatures, near to the optimum value of the ( $p, ^{11}\text{B}$ ) reaction fusion cross-section.

Other important results, arising from the use of the self-consistent multi-fluid code concern: i) The decoupling between the  $p, ^{11}\text{B}$  fusion species temperatures and the electron temperature, which appears simultaneously after the approximation of the critical value of the alpha density (see above paragraph), ii) The continuous temporal rise of the  $p, ^{11}\text{B}$  fusion species temperatures and the simultaneous temporal decrease of the electron temperature and iii) The  $T_B/T_e$  ratio reaches a value, that is higher than 2 ( $T_B/T_e > 2$ ), when the  $p, ^{11}\text{B}$  fusion species temperature and the  $Q = P_{\text{fus}}/P_{\text{Brems}}$  ratio attain their maximum value. The aforementioned results allow the evaluation of: i) The medium conditions, that are related with the achievement of break-even ( $Q = P_{\text{fus}}/P_{\text{Brems}} = 1$ ) and self-sustained fusion ignition conditions ( $Q = P_{\text{fus}}/P_{\text{Brems}} > 1$ ), ii) The necessary time interval for the  $Q > 1$  maximization and iii) The corresponding values of the  $p, ^{11}\text{B}$  fusion species temperatures and alpha density production.

For the examined cases of the ( $p, ^{11}\text{B}$ ) fusion medium, the temperatures of the fusion species decline after the maximization of the ( $P_{\text{fus}}/P_{\text{Brems}}$ ) ratio (figure 3, 4, 5), due to the depletion of the Boron species, the decrease of the alpha energy and the emittance of Bremsstrahlung radiation losses from the remaining density of protons (fusion species with the  $n_p/n_B = 10$  ratio). As the alpha temperature decreases, due to the collisions of the alphas with the medium electrons, less energy is transferred to the  $p, ^{11}\text{B}$  fusion species. Consequently, the  $p, ^{11}\text{B}$  fusion process gradually stops, as well as the alpha density production.

A potential experimental setup for the verification of the conditions related to ignition and self-sustained fusion burn ( $Q = P_{\text{fus}}/P_{\text{Brems}} > 1$ ) of the  $p\text{-}^{11}\text{B}$  medium, could be a mirror like magnetic configuration, with  $p\text{-}^{11}\text{B}$  solid targets (like Borane) at its extremities. The fabrication of the solid targets with the appropriate density ratio ( $n_p/n_B > 1$ ) is under investigation by the H11B Energy Company [49] and remains as an important challenge until today. The laser beam intensity interaction with the solid targets enables the injection of the  $p, ^{11}\text{B}$  fusion species in the magnetic configuration, with specific proton to boron density ratios ( $n_p/n_B > 1$ ) and with various initial densities and energies. The details, concerning the development and the operation of such a compact device, is outside the objectives of the present work. However, our simulations enable the description of the temporal evolution of the physical parameters, as well as the estimation of the medium trapping time interval, which is necessary, in order to obtain the maximum fusion power density output ( $P_{\text{fus}}$ ). For a low density  $p\text{-}^{11}\text{B}$  medium ( $n_p$  between  $10^{20} \text{ m}^{-3}$  to  $10^{21} \text{ m}^{-3}$ ), the medium energy confinement time ( $\tau_E$ ) in a Compact Magnetic Fusion Device, should range between 1 sec and 10 sec.

Also, for all the examined cases, the fusion products consist of only charged particles. Therefore, in a potential compact fusion device with a magnetic trapping field of 4T to 6T, the conversion of the energetic charged fusion species into useful energy could be carried out at the time of their maximum energy, via their extraction from the device, using a magnetic nozzle. Without extraction of the charged particle species, the variation cycle of the beta, from a low value (initial density and temperature of the fusion medium) to a high value (maximum temperatures) and again to a low value (very low medium temperatures  $\sim 0$ ), would enable the formation of a “magnetic piston”, the energy of which could be converted into electricity. Both of these schemes are not yet sufficiently explored or experimentally applied and are out of the objectives of the present work.

An important remark of the presented numerical simulations is that, the initial two fluid ( $p, ^{11}\text{B}$ ) fusion medium with temperature  $T_{\text{in}} = 150 \text{ keV} - 200 \text{ keV}$ , is transformed into a high energy three fluid ( $p, ^{11}\text{B}$  and  $\alpha$ ) medium, after 1 sec of fusion process (figure 5). This fact is very essential for the energy conversion of the charged fusion particles into electricity. For the case of the ( $p, ^{11}\text{B}$ ) fusion medium, with initial proton density:  $n_p = 10^{21} \text{ m}^{-3}$  and  $n_p/n_B = 10$ , the numerical results show a maximum fusion power density production:  $P_{\text{fus}} = 40 \text{ MW/m}^3$  (at the maximum  $Q > 1$  value), confirming that the ( $p, ^{11}\text{B}$ ) fuel, could be a promising candidate for clean energy production.

## A Description and equations of the used zero order (0-D) multi-fluid global particle and energy balance model

The zero-dimension (0-D) multi-fluid global particle and energy balance model, which was used for the performance of our numerical simulations, is based on the time-dependent form of the energy conservation law from fluid dynamics (chapters 4, 9 and 10 of [43]), the species density evolution of (eq. 10.27 of [43]) and the following considerations: i) The medium is composed by four (4) separate fully ionized fluid species ( $p, ^{11}\text{B}, e, \alpha$ ), with Maxwellian temperature distribution functions. ii) The energy density leaving the volume by convection and the energy losses due to expansion and diffusive processes are neglected (only temporal evolution is considered, not spatial). Thus, depending on the initial  $p\text{-}^{11}\text{B}$  fuel density case, the alpha density production and the potential development of the chain reactions alpha heating effect and the related avalanche effect are examined, only for time intervals and acceptable configurations without plasma expansion.



In a small plasma fluid element, of volume  $\Delta V$ , the time rate of change of internal energy (power density), per unit of volume  $\Delta V$ , is [43]:

$$\frac{d}{dt} (\text{internal energy density}) = \frac{d}{dt} \left( \frac{3}{2} k_B n_s T_s \right) = \left( \frac{3}{2} k_B n_s \frac{dT_s}{dt} \right) + \left( \frac{3}{2} k_B T_s \frac{dn_s}{dt} \right) \quad (\text{A.1})$$

In equation (A.1):

- i) The first term on the right side corresponds to the net energy exchange collision rate (temperature equilibration term) of each plasma species  $s$  (p, B, e,  $\alpha$ ), as a consequence of its binary Coulomb collisions with the rest plasma species  $s'$ . In chapter 2 of [45], chapter 9 of [43] and [42, 46], there are relevant and equivalent equations, concerning the net energy exchange collision rate, due to binary Coulomb collisions:

$$\frac{3}{2} n_s \frac{dT_s}{dt} = -\overline{\nu^{s/s'}} n_s (T_s - T_{s'}) \quad (\text{A.2})$$

The used collision frequencies  $\overline{\nu^{s/s'}}$  are from [43, 45], where  $T$  in place of  $k_B T$ .

- ii) The second term on the right hand side of (A.1) denotes the p,  $^{11}\text{B}$  species internal energy density losses, due to nuclear fusion reactions:

$$\frac{3}{2} k_B T_s \left( \frac{dn_s}{dt} \right) = \frac{3}{2} k_B T_s (n_p n_B \langle \sigma v \rangle) \quad (\text{A.3})$$

Where:  $\langle \sigma v \rangle$  is the fusion reactivity for p- $^{11}\text{B}$  reaction. In the context of the model, the p- $^{11}\text{B}$  fusion reactivity calculations exploit the cross section parametric equation of [11] with the improved and/or updated values of [47].

## B Equations of the alpha heating of the multi-fluid model for HB11 fusion

### B.1 Temporal evolution of the fluid species (p, $^{11}\text{B}$ , e, $\alpha$ ) mass densities

$$\frac{d\rho_e}{dt} = 0 \quad (\text{B.1})$$

$$\frac{d\rho_p}{dt} = -m_p S_r \quad (\text{B.2})$$

$$\frac{d\rho_B}{dt} = -m_B S_r \quad (\text{B.3})$$

$$\frac{d\rho_\alpha}{dt} = 3m_\alpha S_r = 12m_p S_r \quad (\text{B.4})$$

Where:

$\rho_i = m_i n_i$ : Mass density of species  $i = \text{e, p, B and } \alpha$  ( $n_i$ : number density of species  $i$ ).

$S_r = n_p n_B \langle \sigma v \rangle$ : p- $^{11}\text{B}$  fusion Reaction Rate in  $\text{m}^{-3} \text{s}^{-1}$ .

$m_B = 11 m_p$ : Approximately the same proton and neutron mass. On this basis:  $m_\alpha = 4 m_p$ .

In (B.1)–(B.4), we neglect terms, concerning the refueling with cold particles.

## B.2 Temporal evolution of the fluid species (p, $^{11}\text{B}$ , e, $\alpha$ ) internal energy densities

Following the definitions of equations A2, A3 from [42, 43], for the power density of each plasma fluid species (p,  $^{11}\text{B}$ , e,  $\alpha$ ), we have  $T$  in place of  $k_B T$  (in the terms referring to collisions):

$$\frac{d\varepsilon_e}{dt} = Q_{ep} - P_{\text{Brems}} = n_e v^{e/\alpha} (T_\alpha - T_e) + n_e v^{e/p} (T_p - T_e) + n_e v^{e/B} (T_B - T_e) - P_{\text{Brems}} \quad (\text{B.5})$$

$$\frac{d\varepsilon_p}{dt} = -Q_{rp} + Q_{pp} = \frac{-S_r k_B T_p}{\gamma - 1} + n_p v^{p/e} (T_e - T_p) + n_p v^{p/\alpha} (T_\alpha - T_p) \quad (\text{B.6})$$

$$\frac{d\varepsilon_B}{dt} = -Q_{rB} + Q_{Bp} = \frac{-S_r k_B T_B}{\gamma - 1} + n_B v^{B/e} (T_e - T_B) + n_B v^{B/\alpha} (T_\alpha - T_B) \quad (\text{B.7})$$

$$\frac{d\varepsilon_\alpha}{dt} = Q_{r\alpha} + Q_{ap} = 3S_r \varepsilon_k - n_\alpha v^{a/e} (T_e - T_\alpha) - n_\alpha v^{a/p} (T_p - T_\alpha) - n_\alpha v^{(a/B)} (T_B - T_\alpha) \quad (\text{B.8})$$

Where:

$\varepsilon_i$ : Internal energy density of species  $i = \text{e, p, B and } \alpha$ .

$P_{\text{Brems}}$ : Bremsstrahlung power density (radiation losses) in  $\text{W/m}^3$ .

$Q_{ep}, Q_{pp}, Q_{Bp}, Q_{ap}$ : Net energy exchange collision rate of each species (e, p, B,  $\alpha$ ), as a consequence of its binary collisions with the rest fusion medium species.

$T_e, T_p, T_B, T_\alpha$ : Temperature of e, p, B and  $\alpha$ .

$Q_{rp}, Q_{rB}, Q_{r\alpha}$ : Change in the internal energy density of p, B reactants and alphas, due to p- $^{11}\text{B}$  fusion reactions.

$\varepsilon_k$ :  $3\varepsilon_k = 8,7 \text{ MeV}$  is the energy of p- $^{11}\text{B}$  fusion produced alphas ( $\varepsilon_k = 2.9 \text{ MeV}$ ).

$v^{s/s'}$ : The collision frequencies, accordingly to the formulations of chapter 2 of [45], for a Maxwellian distribution of test particles of species  $s$  colliding with a Maxwellian distribution of background particles  $s'$ . For the example case of the Coulomb collisions of a proton with an electron of the plasma:

$$n_p v^{p/e} = \frac{4}{\sqrt{\pi}} \frac{4\pi n_p n_e q_p^2 q_e^2 \ln \Lambda_{pe}}{(4\pi \varepsilon_0)^2 m_p m_e u_{Tpe}^3}; \quad \text{similar expressions for all species} \quad (\text{B.9})$$

$u_{Tpe}$ : The appropriate mean thermal velocity for the combination of protons and electrons or alphas (both with Maxwellian distributions):

$$u_{Tpe} = \left[ 2 \left( \frac{T_p}{m_p} + \frac{T_e}{m_e} \right) \right]^{1/2} = \sqrt{u_{Tp}^2 + u_{Te}^2} \quad (\text{B.10})$$

A relative expression of the above equation of (B.9) is the equation 9.118 in chapter 9 of [43], with  $T$  in place of  $k_B T$ .

The numerical solution of the 1<sup>st</sup> order system of differential equations of species mass densities and internal energies densities ((B.5)–(B.8)), as a function of time, enables the evaluation of the temporal evolution of: i) Fusion reactivity  $\langle \sigma v \rangle$ , ii) Reaction rate  $S_r$ , iii) Fusion medium species densities and

internal energies (energies, as well) iii) Fusion power density production, iv) Bremsstrahlung power density losses, v) Ignition criterion value:  $Q = (P_{\text{fus}}/P_{\text{Brems}})$ .

The multi-fluid model could also be adjusted for the case of different initial fuels, such as D-T or D-<sup>3</sup>He, which are used by the private companies of General Fusion [34, 35] and Helion Energy [36], developing fusion energy in compact device.

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